

Andamio Whitepapers

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Andamio Light Paper

Andamio is a credentialing solution. It allows people to issue, earn, and verify credentials that can be carried anywhere.

Andamio means scaffolding. Scaffolding is a temporary structure that helps something stand up for the first time, and then comes down.

Andamio is scaffolding that people use to build trust.

Trust Signals

On the Internet, content is now effectively infinite, but trust signals are scarce.

Here are three questions for you:

1. How do you know who you can trust with your data?
2. How do you know who you can trust to tell the truth?
3. How do you know who you can trust to do what they say they will?

Credentials are how people have answered questions like these for a long time. We think they are due for an upgrade.

Design Decisions

We built Andamio by making a handful of design decisions about how credentials could work differently. The foundation is a public blockchain ledger, infrastructure no single company owns. Every decision below follows from putting credentials there instead of inside a company's database.

1 Credentials are immutable and stored on public infrastructure

No one can quietly change them or switch them off, and they do not disappear when a company does. A credential outlives whoever issued it.

2 Credentials are composable

One credential can be required to earn another. What you earn opens the door to what comes next, so credentials build on each other instead of piling up in a drawer.

3 Credentials are useful

A credential is more than something to display. Software can read its contents, and an application can check that someone holds one and act on it. A credential can gate access, unlock the next step, or drive what an app does, not just sit in a profile.

4 Proof is public and evidence is private

Anyone can verify that a credential is real, while the work behind it stays yours. A diploma is public, but the exam and the answers are not. This is how credentials have worked for centuries, and Andamio keeps it that way. The proof goes on a ledger anyone can read, and everything behind it stays private: the work an individual did to earn the credential, and how an organization grants it.

5 **Anyone can issue a credential**

A credential makes no claim about its own value; value comes from use. Issuance is the public good. Creating meaning is the issuer's responsibility.

6 **The holder owns the credential**

It lives with the person who earned it, not in the issuer's system, and they can carry it anywhere it is useful. Value stays where the work was created.

Learn More

If you want to know what you can put to work today, read [Andamio Issuer](#). If you want to know how it works and how to build on it, read [Building on Andamio](#).

About Us

Andamio is built by a globally distributed team, in the open. The way we govern ourselves is part of the product: the decisions we have agreed to are written where anyone can read them. We named the company for a structure that helps you stand, and then comes down. You keep the credential and the trust you built; the scaffolding gets out of the way.

Andamio Issuer

The problem with badges

You issue credentials, and they need to mean something to whoever relies on them: an employer, a partner, or your own team. But the digital credential most organizations issue is a "badge", and badges are not effective.

A badge is a picture that can be shared on LinkedIn, but it isn't data your systems can act on. In a 2025 survey, 91 percent of employers said they look for digital credentials when they hire, but only 34 percent of the organizations issuing them provide data a hiring system can actually read.¹

Compounding the problem, your badges live in your vendor's database, where they can be changed, switched off, or lost if the vendor closes its doors. The vendor, not you, effectively controls your credentialing system.

Andamio turns your badge into a credential you control. It adds a verifiable credential layer on top of the programs you already run, anchored on a public ledger no single company controls. You get everything a blockchain guarantees and none of the blockchain to learn. No wallets, no tokens, nothing new for you or the people you credential to understand.

Six design decisions make an Andamio credential a building block instead of a badge, and the rest of this paper walks through each one.

What makes an Andamio credential different

1 It outlives whoever issued it

A badge lives in your vendor's system, which means it can be quietly changed, switched off, or lost when a contract ends or a company folds. An Andamio credential is immutable and stored on public infrastructure that no single company owns. It cannot be altered after the fact, and it does not disappear when a vendor does. You are not locked in, and neither are the people you credential.

2 It builds on other credentials

A badge stands on its own, and nothing connects to it after it is earned. An Andamio credential can be required before someone takes on the work that earns another, so what they hold opens the door to what comes next. That requirement is enforced by the blockchain itself, not by an app's rules, which means it holds even across organizations: a credential your program issues can be the prerequisite for a program someone else runs, and the protocol will not let them start that gated work until they hold yours. This already runs on mainnet, across credentials from programs that never coordinated. They build on each other instead of piling up in a drawer.

3 It is something software can act on

A badge is a picture. At best it carries a little metadata, and as the opening numbers show, most issuers do not even provide data a hiring system can read. An Andamio credential is machine-readable and programmable. An application can check that someone holds it and gate access on that, unlock the next step, or drive what the app does next. The credential stops being something you look at and becomes something your systems can use.

4 **Its proof is public; its evidence is private**

More than half of job seekers admit to having lied on a resume,² yet badge verification is mostly theatrical, and the systems that try to do it properly often have to expose the work itself. On Andamio, anyone can verify that a credential is real, while the work behind it stays with the person who earned it.

This is how university credentials already work: a diploma is public; the exam papers are not. The signal goes where anyone can check it, and the evidence stays private, with no surveillance of the work.

5 **You own what it means**

With most platforms, the platform decides what your credential is worth and how it can be used. On Andamio, the barrier to issue is low and the credential makes no claim about its own value. You define what it certifies, and its value comes from the track record it earns once it is out in the world. Andamio provides the issuance infrastructure; the meaning is yours.

6 **The earner keeps it**

A badge is trapped in the system that issued it, which is a large part of why so few are ever seen again. An Andamio credential lives with the person who earned it, not in your system, and they can carry it anywhere it is useful, even if they leave your program. Counterintuitively, that is what makes it worth more: a credential people can actually take with them is one they will actually use.

A New Kind of Badge

These six design decisions are why an Andamio credential works as a building block. It's a new kind of badge that outlives its issuer, builds on others, can be acted on by software, proves itself publicly while protecting the work, means what you say it means, and belongs to the earner.

An Andamio credential is infrastructure that the next person, the next organization, or the next system, can trust without taking anyone's word for it. That is the kind of badge that's ready for the real world, and increasingly, for how the work gets done.

For what each plan includes, see [Andamio's published pricing](#).

Sources

¹ Accredible, *2025 State of Credentialing Report* (survey of 502 HR and recruiting leaders and 175 credential issuers, spring 2025): 91% of employers look for digital credentials when hiring; 34% of issuers provide structured data hiring systems can use. <https://www.accreditable.com/reports/2025-state-of-credentialing-report>

² StandOutCV, *How many people lie on their resume to get a job?* (survey of 1,785 employed Americans): 55–64% report having lied on a resume at least once. Multiple independent surveys (ResumeLab, Resume Builder, 2023–2025) cluster between 44% and 70%. <https://standout-cv.com/usa/stats-usa/study-fake-job-references-resume-lies>

For how this works under the hood, see [Building on Andamio](#). For what Andamio is and why it is built this way, see [the Andamio Light Paper](#).

Building on Andamio

Andamio is a protocol you build on, and it gives you primitives at two levels.

At the base is the **on-chain layer**: a set of Cardano spending validators and minting policies that enforce who can issue a credential, who can review work, and what has to be true before one is earned. A credential here is not a picture and not a database entry; it is a hashed artifact on a public ledger, and the rules live on the blockchain, not in any application's code. You can transact against these contracts directly.

On top sits the **Andamio API**, which wraps blockchain primitives into endpoints you can build applications and systems on, without writing transactions yourself. Most builders work at this level. (Higher still, the Andamio Issuer product abstracts the chain away entirely, no wallets and no tokens; that is the subject of a separate paper.)

The same rails carry people and the agents acting for them. Everything in this paper runs today unless it is marked as roadmap.

1 The loop: Commit, Review, Claim

Everything you build on Andamio runs through one loop, exposed as an API: **Commit, Review, Claim**. If you know git, you already know this pattern. Someone takes on a unit of work, it gets reviewed, and the result is recorded for good.

- **Commit.** An access token holder commits to a piece of work. The task definition is hashed at the moment of commitment, so the on-chain record is what they agreed to do, not what the definition happens to say later. The commitment itself is a token the chain now tracks.
- **Review.** The holder produces evidence: an explanation, a working PR, a transaction hash, a design doc. An authorized reviewer evaluates it against the work's Student Learning Targets, the specific capabilities it is built to develop, and either accepts it or refuses it with specific feedback. The reviewer's authority is granted on-chain and enforced by a validator, so not just anyone can sign off, not even on a course or project you forked.
- **Claim.** When the work passes, the holder claims the credential and it attaches to their identity. The attestation is the explanation: the record carries who committed, what they agreed to, who reviewed it, and when.

These three steps run the same whether the work is a course assignment or a project task, the two systems the protocol ships today (section 2). They differ at the claim: a project funds its tasks, so a contributor who completes one claims a reward along with the credential, having staked nothing of their own.

The loop does not care whether the holder is a person or an agent: the same Commit, Review, Claim surface records "I can" and "it can" the same way. It was built for humans, and the agent world is converging on the shape Andamio already ships.

The work has to come from somewhere, and that somewhere is you. Before anyone commits, an issuer **defines** a course or project, its modules or tasks and the Student Learning Targets they develop, and **publishes** it on-chain through the API (section 3). Once your credential exists, it can become the prerequisite for someone else's work, enforced by the chain rather than by an app (section 2).

2 The on-chain layer: how the rules hold

Protocol-level enforcement is possible because of Cardano's eUTxO model: the rules live in on-chain validators (often referred to as "smart contracts"), not in any application's code, so they hold no matter which app, or which agent, is in front of them.

The pieces, in the protocol's own vocabulary:

- **Access Token.** Each user has one. It is a token pair minted under a single policy and named for a self-chosen alias: a **U token** held in your wallet as proof of ownership, and a **G token** held at the global-state address, whose datum carries your record of participation across every organization you touch. One address holds every user's G token, so identity lives at a predictable on-chain location while you keep control through the wallet-held U token. Identity is portable by default, not per-app.
- **Local State.** Where the work lives. The protocol ships two systems today, **Course** and **Project**, and each can have many **instances** owned by individual access token holders ("Intro to Woodworking" owned by Alice, "Contribute to Andamio" owned by Carol).
- **Global State.** The registry and index over all local states: the access token system, protocol governance, and the record of which instances are authorized to issue credentials.
- **Validators.** A set of on-chain scripts (written in Plutus and Aiken) enforce the rules: a local-state validator (Course and Project are instances of it), a global-state validator, the index that ties them together, and the minting policies for access tokens and for state.

The on-chain datum stores only `PolicyId` → hash pairs as small verifiable fingerprints, and any validator can read any user's global state. That is what lets a credential earned under one policy be required by another, enforced by the chain rather than by an app, while keeping proprietary data private. This is impossible in centralized services where competing credentials are illegible to each other. On Andamio, no one has to be trusted, because the rule lives in the validator.

A project can declare that a credential from another issuer is a prerequisite, and the check fires at commit time: when a holder goes to commit to the gated work, the validator confirms their global state already contains the required credential before it will let the commitment mint. This runs on mainnet today, across credentials issued by programs that never coordinated. (The portable, human-facing verifier that reads those prerequisites out of a credential for you is still in flight; more in section 4.)

3 Build on the API

The quickest way to see Andamio is to use it: app.andamio.io is the live app, open to anyone, where you can take a course, commit to work, and earn a credential right away. It runs entirely on the Andamio API, the same one you build on.

The Andamio API (api.andamio.io) turns the on-chain primitives into endpoints. You query it for data and for valid transactions, and the blockchain enforces the rules underneath. Its full reference is generated from the spec, `andamio-dev` plugs into your agent and walks you through it, and the documentation is at docs.andamio.io.

Create your own credentialing app

Use the API to create a new course or project. You get back a course ID or project ID that is yours and published on-chain, and you are now an issuer: you define what the credentials mean, and every credential earned against them carries your policy. From there you build whatever application you want around that ID. The fastest start is `andamio-app-template`, which lets you build from the same UX as app.andamio.io, pointed at your own courses and projects.

Build on what already exists

You do not have to own a course or project to build on one. Given access to the data, the same API lets you build on courses and projects other issuers have created: a dashboard over a partner's credentials, a verifier, a directory, or an agent that commits to work on someone's behalf. Because credentials are public and composable by design, building on another issuer's work is a first-class use, not a workaround.

4 Privacy and verification

Andamio anchors trust without leaking data. The rule is simple: private data cannot go on a public blockchain, so Andamio does not put it there. It puts the hash. A Blake2b hash of the off-chain evidence goes on-chain; the evidence itself stays with whoever produced it. Anyone can confirm a credential is real; no one has to expose the work behind it.

The verification model is fetch, recompute, compare: read the on-chain anchor, recompute the hash of the artifact you hold, and check that they match. You do not have to trust Andamio; you can recompute the hashes yourself, and the compiled bytes are already public on-chain, so there is nothing to wait for.

5 Use your agent

Andamio's developer surface is small enough to hand to your agent. Point it at the **andamio-dev** repo (from section 3), the knowledge layer that teaches it the protocol and estimates costs, and it drives the **Andamio CLI** to build, sign, and submit transactions for you.

The CLI ([andamio-cli](#), v0.13.3) wraps the full transaction lifecycle, build, sign, submit, register, and confirm on-chain, and is the source of truth for what an Andamio transaction looks like. It calls the API at `preprod.api.andamio.io` for development and `mainnet.api.andamio.io` in production. An access token is the one prerequisite for committing work or earning a credential, and `andamio-dev` walks you through getting one.

6 Future development

Easy Onramps

Earning a credential on-chain still asks an end user for a Cardano wallet and an access token. That is the last hard edge in the experience, and smoothing it is next on Andamio's roadmap.

The lever is **sponsored transactions**: on the enterprise tier an issuer covers all user-facing transaction costs, so the people you credential never hold ADA or manage a wallet. Billing is ordinary; the chain is invisible to them. This path is built and exercised on preprod today, with an issuer-funded sidecar signing each user's transactions; mainnet activation is the step that remains.

It is coming soon, not fully delivered. If you are building on Andamio now, you can be part of that journey: develop against the sponsored path on Preprod, and you will be ready the day it goes live on Mainnet.

New Local States

"Course" and "Project" are not the entire Andamio protocol. They are the first two systems built on it. Both are instances of one `local-state` pattern that plugs into the shared `global-state`, which is why a credential earned in either composes with everything else.

The same machinery, Commit, Review, Claim, anchored by a hash and enforced by a validator, can carry any such loop you can define. Building an entirely new local-state system is protocol-level work, not a turnkey

feature, but it is the deepest sense in which Andamio is a protocol you build on: courses and projects are where Andamio started, not the edge of what the pattern can hold.

For what Andamio is and why it is built this way, read the Andamio Light Paper. For what you can put to work today and what it saves you, read Andamio Issuer. This paper is the how.